INTEL Vs AMD

The good old battle of yore, which continues even today, though on a smaller scale and with less fervour

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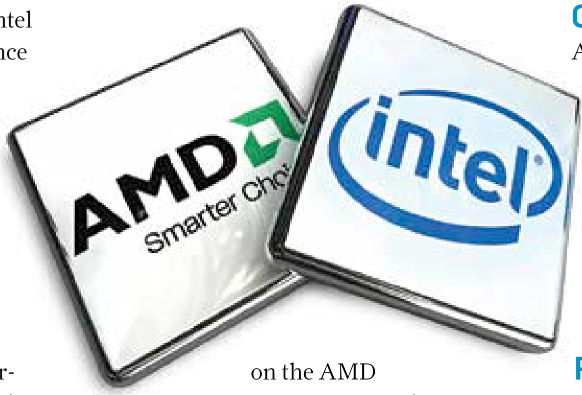
The endless bickering between Intel and AMD has been going on since time immemorial. We're going to be looking at a few more facets than just performance crowns, however.



Performance

Let's get this out of the way. When it comes to high-end PCs AMD doesn't even feature, be it gaming or pure workstation performance. While over-clocking records are broken on a regular basis by AMD processors the winner in a performance showdown almost always happens to be an Intel processor.

Score: Intel 1 - AMD 0

R&D

Intel was one of the top four big-spenders in R&D last year. They spent close to 19% of their revenue on R&D, and though AMD also put down 20% of their revenue for R&D, that works out to $10.1 billion for Intel and $1.2 billion for AMD in 2013. Since we don't award sympathy points for underdogs, this one goes Intel's way too.

Score: Intel 2 - AMD 0

On-die GPU

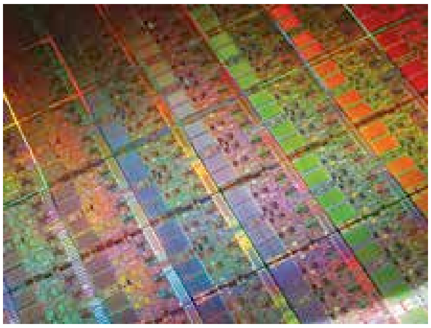
Intel doesn't even stand a chance in this arena, AMD bought out ATI and that gave them an edge over Intel that could've been easily matched had Intel and NVIDIA not had a falling out back in 2009. They had a cross-licensing agreement in place much earlier than AMD's acquisition of ATI. So even with plenty of time to build a good lead, things didn't quite fall in Intel's favour and we can now see the result with AMD's processors performing way better than Intel when it comes to on-die GPU performance. In Bioshock Infinite, the HD 8670D found on the AMD A10-6800K provides playable frame rates at medium configuration while Intel's HD4600 on the i5-4430 can't even manage that.

Score: Intel 2 - AMD 1

Getting your money's worth

AMD's flagship processors might not perform as fast as Intel's, however for anything but the highest budgets, AMD CPUs offer better performance to price ratios. And this is not just with the processor in focus, AMD's on-die GPU scales well with AMD discrete GPUs which is something Intel hasn't been able to replicate mainly due to licensing issues. While Virtu was something Intel tried out with the Sandy Bridge / Ivy Bridge it hasn't quite caught on just yet.

Score: Intel 2 - AMD 2



Ah silicon, where would we be without you?

Console wars

Again, Intel is not even present here. The Xbox 360 had a AMD Xenos processor, the Wii featured an ATI (now AMD) Hollywood processor and the PS3 had an NVIDIA RSX processor handling graphics. Upcoming consoles also feature custom AMD APUs so... victory AMD.

Score: Intel 2 - AMD 3

Power efficiency

Intel's has had the upper hand in power efficiency and they've managed to maintain that edge over the years. And given the tick-tock release cycle they have, the processors get more and more efficient with each passing generation. Finally, the 3D transistor technology Intel adopted recently has even furthered this lead.

Score: Intel3 - AMD 3

Sockets

Over the past six years, Intel has come out with five new sockets while AMD had four. A blatant way of jumping sales (not just for processors but motherboards also)? There is a way to avoid this by having the chipset embedded on the processor itself. AMD's Kabini and Intel's Atom SoCs are doing just that except they will be limited to the laptop/tablet market.

Score: Intel 4 - AMD 4 (this one's a tie)

The scores are tied because of AMD's diversification and also because it was recently chosen for both the new consoles. However, it must be noted that AMD only recently started making a profit from the consoles, while Intel has rarely had a bad year. With sheer volume of sales in hand and better performing products overall, Intel seems to be the winner, tied score or not!